Computer Networks Lab

Cisco Packet Tracer — Assignment 7

All practicals must be completed inside **Cisco Packet Tracer**.
Each assignment requires exactly **two screenshots** as submission.

Topic: Dynamic Host Control Protocol

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_Set7.pkt` (e.g. `20221234_Set7.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of the set.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

Harbor View Clinic has a front-desk area with three workstations used by reception staff. Dr. Anika Verma, the clinic's IT administrator, wants every workstation to receive its IP configuration automatically from the router, so that nobody has to visit a desk to fix settings by hand.

Recently, a visiting contractor plugged in an unauthorised address server. It began handing out addresses from its own range, with a wrong default gateway, and several PCs lost their internet access. Dr. Verma asks you to first reproduce this problem and then stop it, so that PCs only accept addresses from the legitimate router.

Device	IP Address	Address Source
Router Gi0/0 (LAN)	192.168.50.1	Static
PC1, PC2, PC3	Assigned automatically	Automatic
Unknown-Server	192.168.50.250	Static (unauthorised)

Subnet Mask: `255.255.255.0` on all devices

Address Pool Parameter	Value
Router pool range	192.168.50.100 – 192.168.50.199
Default gateway (router pool)	192.168.50.1
Unknown-Server pool range	192.168.50.201 – 192.168.50.210
Unknown-Server default gateway	192.168.50.254

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View with all devices connected and IP labels visible, along with the Command Prompt of the PC that accepted the false address, running `ipconfig` and showing the address from 192.168.50.201 – 192.168.50.210.
2. The switch's CLI output showing the protection feature you enabled, placed side by side with the Command Prompt of any PC running `ipconfig` after the final Release and Renew, showing an address from 192.168.50.100 – 192.168.50.199.

Note: If a PC does not receive an address, first confirm that the router's Gi0/0 interface is **no shut** (active) and that the switch port is up. After making any change, always perform a **Release** and **Renew** on each PC, because existing addresses are not re-checked automatically. Refer the Updated guide on Github Repository to learn more about DHCP Snooping.